



NITI Aayog



SECTOR AT A GLANCE 14

Negative Emission Technologies

Energy in India

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NEGATIVE EMISSION TECHNOLOGIES

ENERGY IN INDIA

The **‘Sector-at-a-Glance’** provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

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Overview

Negative emission technologies (NETs) refer to innovative approaches and technologies that actively remove greenhouse gases – usually carbon dioxide – from the atmosphere. The process is referred to as Carbon Dioxide Removal (CDR) if it involves removing carbon dioxide, or more broadly, as Greenhouse Gas Removal (GGR) if it involves removing gases other than CO₂. These methods include both nature-based solutions, such as afforestation, reforestation, soil carbon sequestration, enhanced weathering and similar processes that leverage natural ecosystems to capture and store carbon dioxide and technology-based solutions, such as biochar, bioenergy with carbon capture and storage (BECCS), direct air capture, and carbon capture, utilisation, and storage (CCUS). In the last ten years, India's CO₂ emissions have consistently increased at a CAGR of about 5%.¹

India has the opportunity to accelerate decarbonization across energy-intensive sectors such as electricity, mining, and steel by incentivizing the large-scale adoption of NETs and meeting its Nationally Determined Contributions (NDCs), particularly the goal of creating an additional carbon sink of 2.5 to 3 billion tonnes of CO₂.²

However, challenges such as the high cost of NETs, limited clear guidelines on adoption of CCUS, limited incentives for private sector investment, and limited workforce capabilities inhibit India's decarbonisation potential.

Given these opportunities and challenges, promoting CCUS adoption in hard-to-abate sectors such as Heavy Commercial Vehicles, Shipping, Steel, Cement, Aluminium, and thermal power plants and accelerating R&D efforts in India's CCUS are the major policy focus areas. Several ongoing and recent central government initiatives, including but not limited to the Draft 2030 Roadmap for CCUS for Upstream E&P Companies focus on promoting adoption and R&D in oil and gas sectors. These efforts, along with several state- and local-level policies and initiatives, seek to accelerate efforts towards decarbonisation.

Promoting the adoption of NETs through policy incentives and financing support, boosting R&D and technology transfer, and enabling greater private sector participation will be critical to continue advancing India's journey on negative emission technologies. This note presents key trends related to different components of negative emission technologies in India and actions for the way forward.

Figure 1: Key negative emission technologies statistics

50%
Reduction in CO₂ emissions aimed by 2050 by GoI

30%
Of all capturable emissions (i.e. 2400 mtpa) aimed to be captured by CCUS technologies by 2050 by GoI

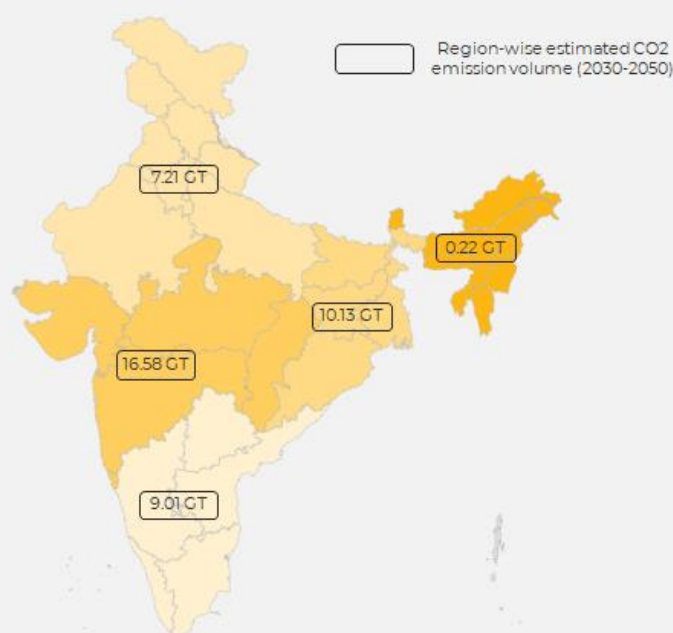
Source: NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021

¹ World Bank, CO₂ emissions (kt) – India, accessed June 2023

² NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021; Cabinet, Cabinet approves India's Updated Nationally Determined Contribution to be communicated to the United Nations Framework Convention on Climate Change, 2022

Figure 2: Region-wise carbon storage clusters in India

Total theoretical storage capacity of India = 395 - 614 giga tonne CO₂



Source: NITI Aayog, Carbon Capture, Utilisation and Storage, 2022

Figure 3: Carbon capture volume in two scenarios and tentative aggregative investment in India

Industry	CO ₂ Emissions in 2030, mtpa	CO ₂ Capture in Base Case, mtpa	CO ₂ Capture in Optimistic Case, mtpa
Gasification based production	27	4	10
SMR/ATR based H ₂ production	102	2	2.5
Cement	325	2	2
Iron and Steel	450	4	5
Refinery and Chemical	177	-	0.5
Coal-based power	1,210	8	15
Total	2,291	20	35
Investment Required	BB\$	4	6.7
	Rs. crores	30,000	50,000

Note:

- For the gasification industry, carbon capture is part of the process. The high concentration CO₂ stream has to be purified and compressed for further disposition. Hence, only the capital cost of CO₂ purification and compression has been considered.
- Delivery pressure of product CO₂ is considered as 100 bar (a) for all cases.
- The capital costs have been calculated considering the capital cost for similar projects in different parts of the world and adapting the same to Indian costs and conditions.

Source: NITI Aayog, Carbon Capture, Utilisation and Storage, 2022

Opportunities and challenges

Opportunities

This section highlights the outcomes policymakers can achieve across sectors by incentivising large-scale adoption of NETs. The Indian government is actively exploring CCUS technologies for various industries that contribute significantly to the nation's economy.



Decarbonisation of energy-intensive sectors:

India relies heavily on fossil fuels to power critical industries such as steel, cement, oil and gas, and chemicals. CCUS technologies allow for the scalable and profitable conversion of waste gases from steel plants, including blast furnaces, basic oxygen furnaces, and coke ovens, into blue hydrogen at less than INR 100 per kg. Moreover, CCUS can enable the production of synthetic fuels like green methanol, sustainable aviation/marine fuel and significantly decarbonise thermal power plants, the largest CO₂ emitters in India, and India's coal gasification economy.³ Captured CO₂ offers versatile use cases that further promote circularity, including producing green urea for fertilizers, carbonating beverages, or packaging food to extend shelf life, converting it into methanol, ethanol, and polymers, and enhancing oil recovery.⁴



Accelerating development and adoption of CCUS tech:

CCUS technologies are being rapidly developed worldwide. NITI Aayog highlights that assimilating and transferring such proven technologies to India's context can reduce technological risks and catalyse domestic adoption.⁵ Additional support and incentives such as cash subsidies for capturing CO₂ and de-risking CCUS projects, viability gap funding, tax breaks, and R&D support for developing and demonstrating novel technological solutions can reduce the cost of using NETs and accelerate the widespread adoption of CCUS-based carbon capture solutions.⁶



Leveraging carbon markets to increase adoption of CCUS:

The Ministry of Petroleum and Natural Gas (MoPNG) has highlighted that carbon markets can act as vital mechanisms to increase the industrial adoption of carbon mitigation methods and technologies such as CCUS, along with enabling CO₂ emission reduction. However, India's carbon markets are still nascent and need to be developed cost-effectively to be widely adopted.⁷



Global leadership in CCUS technology:

Given the global nascency of CCUS technologies, India has the unique opportunity to emerge as a global innovation hub in low-cost carbon capture methods and technologies. India has access to a vast spread of onshore and offshore sedimentary basins that can provide long-term carbon storage solutions.⁸ Over the next decade, investments in R&D, business incubators and collaborations with universities and foreign businesses to develop CCUS technologies can elevate India's role in global NET markets and present a good long term revenue generation opportunity for the CCUS industry.⁹

3 M. N. Dastur & Company and NITI Aayog, Carbon Capture, Utilization and Storage (CCUS), 2022, Ministry of Petroleum and Natural Gas, Carbon Capture Utilization & Storage, accessed June 2023

4 M. N. Dastur & Company and NITI Aayog, Carbon Capture, Utilization and Storage (CCUS), 2022, Ministry of Petroleum and Natural Gas, Carbon Capture Utilization & Storage, accessed June 2023

5 NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021

6 NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021

7 Ministry of Petroleum, Oil and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022

8 Ministry of Petroleum, Oil and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022

9 Ministry of Petroleum, Oil and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022



Biochar can help in climate-mitigation:

Biochar is a carbon-rich substance derived from residual waste of renewable sources, including bamboo, corn cobs, cotton stalks, lantana, mulberry, and other agricultural and woody biomass. According to a study conducted by IIT Delhi, biochar has the potential to sequester approximately 376.11 ± 78.52 million metric tons of CO₂, presenting an opportunity for India to reduce emissions from agricultural and allied activities by 41.41–63.26%, as indicated by the research findings.¹⁰

Challenges

This section highlights some of the major challenges and constraints for negative emission technologies in India:



High cost of NETs:

Since NETs are still in their early stages of development, the cost of adopting these technologies is high and the time required to set up the infrastructure is substantial. Even within CCUS technologies, the costs can vary significantly for different applications. Capturing CO₂ directly from the air is currently the most expensive approach.¹¹ When capturing CO₂ from industrial processes with concentrated CO₂ streams, like ethanol production or natural gas processing, the cost is approximately INR 1000 - 2000 per ton of CO₂. The cost is higher for processes with more diluted gas streams, such as cement production and power generation, ranging from about INR 3000-10,000 per ton of CO₂.¹²



Need for policy frameworks and carbon markets:

There are limited clear guidelines and incentives to promote the adoption of CCUS technologies across various sectors via R&D, awareness campaigns, development of critical infrastructure etc. Without a supportive policy environment, industries have limited motivation to invest in CCUS projects and incorporate them into their operations. Moreover, the absence of regulatory mechanisms for monitoring, reporting, and verifying CO₂ storage and utilisation activities hinders the implementation of CCUS in India.¹³ Further, though carbon markets can be a potent catalyst in promoting CCUS usage, India is still in the early stages of developing the Indian Carbon Market (ICM).¹⁴



Need for more private sector involvement:

The Ministry of Petroleum and Natural Gas notes that private sector contribution in areas such as R&D and demonstration of carbon capture capabilities is essential to prove the economic viability of CCUS technologies and increase their economy-wide uptake.¹⁵ However, there are currently limited incentives to attract their support, such as subsidies, targeted funding for capital and operating costs, tax credits, and viability gap funding for commercial projects.¹⁶



Limited workforce capabilities:

There are little to no professional courses on CCUS in Indian universities, R&D institutes, and vocational education institutions, leading to the lack of a qualified workforce that can accelerate India's decarbonisation efforts.¹⁷

¹⁰ Abhijeet Anand, Vivek Kumar, Priyanka Kaushal, Biochar and its twin benefits: Crop residue management and climate change mitigation in India, Renewable and Sustainable Energy Reviews, Volume 156, 2022,

¹¹ NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021

¹² International Energy Agency, Is carbon capture too expensive?, 2021

¹³ Ministry of Petroleum and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, accessed June 2023

¹⁴ Ministry of Power, Ministry of Power & Ministry of Environment, Forests & Climate Change to develop Carbon Credit Trading Scheme for Decarbonisation, 2023

¹⁵ Ministry of Petroleum, Oil and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022

¹⁶ Ministry of Petroleum, Oil and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022

¹⁷ Ministry of Petroleum, Oil and Natural Gas, Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022

Institutional framework and policy landscape –

Key ministries and departments

This section highlights the major central government bodies and national-level institutions that oversee solar energy.



NITI Aayog

Conducts policy analysis of energy sectors related to power, coal, petroleum and natural gas, new and renewable energy, and atomic power; and undertakes steps to improve energy planning, and data management in India. It has played a crucial role in establishing a framework to promote CCUS technologies, facilitate the development of CCUS policies, and create a viable economic model for CCUS adoption and implementation across various industries and sectors.



Ministry of New and Renewable Energy (MNRE)

Develops renewable and clean energy technologies, systems, products, and services in line with global standards and deploys these for energy applications for sectors like power, transport and support country's objective of achieving energy security



Indian Renewable Energy Development Agency

Finances and supports self-sustainable investments in energy generation from renewable sources, and carry out promotion of energy efficiency and sustainable environmental technologies



Department of Science and Technology

Formulates policies, promotes scientific research, facilitates technology transfer along with international collaboration, and capacity building for societal development

Major policies and programs

This section highlights existing policies related to flood management and their most prominent features. A more comprehensive list of policies is provided in the policy repository of the NITI Knowledge Platform.



Draft 2030 Roadmap for Carbon Capture Utilization and Storage (CCUS) for Upstream E&P Companies, 2022¹⁸:

The MoPNG is undertaking initiatives to promote the development and adoption of CUSS and Carbon Capture and Storage (CCS) technologies in the country's oil and gas sector for Enhanced Oil Recovery (EOR) from mature fields. Additionally, a Memorandum of Understanding (MoU) has been signed between the Ministry and IIT Bombay to establish a practical framework that will accelerate research and development efforts in India's carbon capture, utilisation, and storage.



¹⁸ Ministry of Petroleum and Natural Gas, Draft 2030 Roadmap for CCUS, 2022



CCUS Policy Framework and its Deployment Mechanism in India, 2021¹⁹:

NITI Aayog developed a policy framework for upcoming CCUS policies in India that outlines incentives to achieve an economic model for CCUS use and encourage adoption in multiple industries and sectors, such as steel, natural gas, and fertilisers. It will act as a critical document that helps different ministries and states create the right policy environment to catalyse decarbonisation.



Inter-ministerial Technical Committees to support CCUS²⁰:

Given that CCUS technologies are still at a conceptual stage, NITI Aayog has established four Inter-Ministerial Technical Committees tasked with offering their expertise and guidance in several areas including the development of safety and technical standards, the capture of carbon, the utilisation of carbon, and the transportation and storage of carbon.



International collaboration and initiatives:

India has engaged in numerous global partnerships to enhance knowledge, develop capabilities, and carry out collaborative research efforts in the field of CCUS technologies. The Department of Science & Technology (DST) plays a significant role in the Mission Innovation (MI) Programme, a global initiative that includes 24 countries and the European Union. This programme has identified CCUS as a crucial area of innovation. To date, the DST has financed 19 projects related to CCUS research and development, collaborating with 13 different countries.²¹ India has also forged partnerships with the US and the UK to advance the development of CCUS under the India-USA Strategic Clean Energy Partnership and India UK 2030 Roadmap, respectively.²²

¹⁹ NITI Aayog, CCUS Policy Framework and its Deployment Mechanism in India, 2021

²⁰ Ministry of Petroleum and Natural Gas, CO₂ Capture Hubs, 2023

²¹ Ministry of Science and Technology, [DST encourages translational research on Carbon Capture, Utilisation & Storage \(CCUS\)](#), 2020

²² Department of Science and Technology, [CCUS in India and the UK](#), 2022; Ministry of Petroleum and Natural Gas, [India-USA Strategic Clean Energy Partnership Ministerial Joint Statement](#), 2023

Way forward

This section highlights considerations for policymakers seeking to advance policy formulation and implementation in this sector. These are based on reports and materials prepared by key government bodies and expert institutions like the NITI Aayog, the Ministry of Coal, and the Ministry of Petroleum and Natural Gas.

 Promote adoption through policy incentives and financing support	Establish an incentive-based policy framework - A comprehensive policy framework that provides incentives for companies to adopt carbon capture technologies is crucial to encourage more businesses to invest in these technologies Fostering public-private collaborations and access to green finance instruments for early movers - Private investment in carbon projects can be attracted by encouraging public-private partnerships. This can be done by providing regulatory support and risk guarantees for private sector investments and new forms of capital such as green bonds Promote emission-reduction strategies in public-sector undertakings - Public sector oil and gas companies can be encouraged to embrace emission-reduction strategies by setting clear emissions targets and providing support for their efforts. This can help to align their operations with national climate goals and create strong market signals for private-sector projects
 Boost R&D and technology transfer	Cultivate international research partnerships and technology exchanges - R&D efforts in carbon capture technologies can be accelerated by fostering international collaborations and partnerships through bilateral or multilateral agreements. This can be achieved through joint research initiatives, shared funding mechanisms, and institutional frameworks for technology transfer Providing support for technological advancements - Safe and cost-effective carbon capture technologies can be developed by providing research grants and funding for innovation. This can help to reduce the costs and risks associated with these technologies.
 Enable greater private sector participation	Create a business-friendly regulatory environment - A conducive environment for private sector participation can be created by providing regulatory certainty and market-based incentives. This can encourage more private companies to invest in carbon capture projects Offering incentives for sustainable practices - Private companies can be incentivized to trap, recycle, and store their emissions through mechanisms such as carbon pricing or emissions trading schemes. This can help to internalize the cost of carbon emissions and promote sustainable practice Encouraging pilot projects in industry - Private firms in the cement and steel industry can be encouraged to pilot carbon capture technologies by providing technical support and recognition for their efforts. This can help to showcase the viability of these technologies and inspire other companies to follow suit

Recommended readings

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